VIT BHOPAL • CSE2002 & TECHNICAL PLACEMENT ROADMAP

Detailed DSA Master Execution Timetable

Start Date: Nov 14, 2026 (Recursion)
CSE2002 Core End: Feb 28, 2027
Advanced End: May 12, 2027

> **Two-Tier Strategic Architecture:** Starting on **November 14, 2026** directly with **Recursion Basics** (with OOPs & Complexity already covered or integrated via your CSE2006 Java course), your curriculum is split into two distinct phases. **Part A (Nov 14 – Feb 28)** completes 100% of your **CSE2002 college curriculum** (Units 1 through 5: Recursion, Divide & Conquer, ArrayLists, Backtracking, Linked Lists, Stacks, Queues, Trees, BSTs, Hashing, Heaps, and Graphs) with locked Zero-DSA buffers during **CAT-1 (Dec 13–23)**, **Jan 27–31**, and **CAT-2 (Feb 03–13)**. **Part B (Mar 01 – May 12)** transitions into university TEE revision followed by advanced interview paradigms.

Part A: CSE2002 College Syllabus Core (Nov 14, 2026 – Feb 28, 2027)

Dates	Days Allotted	Syllabus Module	Topics Covered & CSE2002 Unit Alignment
Nov 14 – Nov 22	**9 Days** 7 Learn + 2 Rev	**22 & 23. Recursion Basics (Part 1 & Part 2)**	Call stack frame mechanics, base cases, recursion trees, subsets, permutations, tiling problem, Tower of Hanoi, plus Big-O recurrence analysis (CSE2002 Unit 1).
Nov 23 – Nov 29	**7 Days** 5 Learn + 2 Rev	**24. Divide & Conquer**	Merge Sort, Quick Sort, rotated sorted array search, and inversion count analysis (CSE2002 Unit 2 Sorting & Searching).
Nov 30 – Dec 03	**4 Days** 3 Learn + 1 Rev	**27. ArrayLists**	Dynamic array resizing, multi-dimensional lists, two-pointer techniques, and Container With Most Water (CSE2002 Unit 1).
Dec 04 – Dec 12	**9 Days** 7 Learn + 2 Rev	**26. Backtracking**	State-space decision trees, branch pruning, N-Queens problem, Grid Ways, and Sudoku solver implementation.
Dec 13 – Dec 23	**11 Days**	**CAT-1 GPA Guard Buffer & Exams**	**Zero DSA Tasks: Dec 13–14 Pre-Exam Buffer + Dec 15–23 CAT-1 Exams. 100% focus on Units 1 & 2 across all 4th-semester subjects.**
Dec 24 – Dec 31	**8 Days** 5 Learn + 2 Rev + 1 Rest	**28 & 29. Linked Lists (Part 1 & Part 2)**	Singly, Doubly, and Circular Linked Lists, in-place reversals, Floyd's cycle detection, and Merge Sort on Linked Lists (CSE2002 Unit 3; Dec 25 holiday rest).

Dates	Days Allotted	Syllabus Module	Topics Covered & CSE2002 Unit Alignment
Jan 01 – Jan 06	**6 Days** 4 Learn + 2 Rev	**30 & 31. Stacks (Part 1 & Part 2)**	Array and Linked List stack implementations, Infix to Postfix conversion, Stock Span, Next Greater Element, and Histogram Area (CSE2002 Unit 3).
Jan 07 – Jan 13	**7 Days** 4 Learn + 3 Light	**32. Queues**	Circular queues, Deques, Queue using two stacks, and stream processing (CSE2002 Unit 3; bridges into Pongal Break for light practice).
Jan 14 – Jan 20	**7 Days** 5 Learn + 2 Rev	**34, 35 & 36. Binary Trees (Part 1, 2 & 3)**	Tree terminologies, Preorder/Inorder/Postorder/Level-order traversals, Height, Diameter, Lowest Common Ancestor, and Top/Bottom views (CSE2002 Unit 4).
Jan 21 – Jan 24	**4 Days** 3 Learn + 1 Rev	**37 & 38. Binary Search Trees (Part 1 & 2)**	BST search, insertion, deletion, BST validation, balanced BST construction, and AVL/Red-Black tree rotation concepts (CSE2002 Unit 4).
Jan 25 – Jan 26	**2 Days** 2 Intensive	**40. Hashing**	Hash functions, Separate Chaining, Open Addressing collision resolution, HashMap/HashSet internals, and subarray sum problems (CSE2002 Unit 4).
Jan 27 – Jan 31	**5 Days**	**Dedicated No-DSA Buffer Window**	**Zero DSA Tasks: 5 full days reserved for clearing Digital Assignments, Lab FAT records, and mid-semester college coursework catchup.**
Feb 01 – Feb 02	**2 Days** 2 Review	**Unit 3 & 4 DSA Consolidation**	Rapid code walkthroughs and complexity revision for Linked Lists, Stacks, Queues, Trees, BSTs, and Hashing prior to CAT-2 lockout.
Feb 03 – Feb 13	**11 Days**	**CAT-2 GPA Guard Buffer & Exams**	**Zero DSA Tasks: Feb 03–04 Pre-Exam Buffer + Feb 05–13 CAT-2 Exams. 100% focus on Units 3 & 4 across all university courses.**
Feb 14 – Feb 18	**5 Days** 3 Learn + 2 Rev	**39. Heaps**	Binary Min/Max Heaps, Heapify algorithm, Heap Sort, Priority Queues, and K-closest/sliding window patterns (CSE2002 Unit 5 prerequisite).
Feb 19 – Feb 28	**10 Days** 7 Learn + 3 Rev	**42–47. Graphs (Part 1–5 & Supplemental)**	Adjacency Matrix/List, BFS, DFS, Cycle Detection, Topological Sort, Dijkstra's Shortest Path, Bellman-Ford, Prim's/Kruskal's MST, and Disjoint Set Union (CSE2002 Unit 5).
Feb 28, 2027	**Target Hit**	**CSE2002 Core Syllabus Complete**	**100% of Units 1–5 of your college DSA syllabus and all foundational data structures are finished on schedule.**

Part B: TEE Buffer & Post-February Advanced Extension (Mar 01 – May 12, 2027)

March is dedicated to university Term End Examinations (TEE). Immediately after finals conclude, you complete the remaining interview-dominant algorithmic paradigms.

Dates	Days Allotted	Syllabus Module	Topics Covered & Placement Preparation Focus
Mar 01 – Mar 31	**31 Days**	**TEE Prep Buffer & Final Exams**	**Zero New DSA Modules: Mar 01–10 Final Lab FAT & 5-Unit Theory Consolidation + Mar 11–31 Term End Examinations (100% CGPA Protection).**
Apr 01 – Apr 06	**6 Days** 4 Learn + 2 Rev	**33. Greedy Algorithms**	Activity Selection, Fractional Knapsack, Job Sequencing with deadlines, Indian Coins, and interval scheduling heuristics.
Apr 07 – Apr 12	**6 Days** 4 Learn + 2 Rev	**41. Tries**	Prefix tree node design, insertion, search, startsWith queries, Word Break problem, and Longest Word with all prefixes.
Apr 13 – May 04	**22 Days** 16 Learn + 6 Rev	**48–53. Dynamic Programming (Part 1–6)**	Memoization vs. Tabulation, 0-1 Knapsack, Unbounded Knapsack, Longest Common Subsequence (LCS), Longest Increasing Subsequence (LIS), Catalan Numbers, and Matrix Chain Multiplication.
May 05 – May 11	**7 Days** 5 Learn + 2 Rev	**54. Segment Trees**	Range sum/min/max queries, point updates, and lazy propagation mechanisms for competitive coding rounds.
May 12, 2027	**1 Day**	**55. Certificate of Completion**	**Full curriculum wrap-up, mock interview self-assessment, and transition into daily LeetCode contest drills for summer internships.**